

1.1 Sound Hardware

When it comes to the basic PC hardware for sound, you, of course, have two options—onboard sound and dedicated sound cards. We talk about both these.

1.1.1 Onboard Sound

Onboard Sound has improved in a big way since the 2-channel solutions of a few years back. All current-gen motherboards come with at least 6-channel sound as a *de facto* standard, with the latest chipsets from the big guns such as Intel and NVIDIA actually having 8-channel audio. For example, Intel released “Azalia,” which is the code-name for its latest 7.1, high-definition audio specification. Other features of Azalia or Intel High Definition Audio include multi-streaming capabilities, which gives you the ability to send two or more different audio streams to different locations at the same time, from the same PC. This means you can play a game’s sounds through your 5.1 speakers while simultaneously voice-chatting on the Net! Also, there is “jack retasking”—one jack can have more than one function. The motherboard can sense when a device is plugged into an audio jack, determine what kind of device it is, and change the port function if the device has been plugged into the wrong port!

Most mid-range to high-end Intel motherboards from the 925X series and up sport Azalia, although some boards use lower audio solutions, mainly due to cost constraints. Azalia is capable of support for up to eight channels at 192 kHz/32-bit quality.

The ALC 880 and ALC 880D are also 7.1-channel high-definition codecs providing four 24-bit, 2-channel DACs. This standard achieves 100 dB sound quality, making these codecs suitable for high-end multimedia PCs.

SoundMAX is a combination of Analog Devices’ AC’97 CODEC series and software that enables your computer to have features that, according to Analog Devices, can even “surpass the functionality of premium soundcards at a fraction of the cost.”